

METABOLISM OF PYRIMIDINE NUCLEOTIDES

INTRODUCTION:

The pyrimidine nucleotides are:

- 1) Cytidine monophosphate CMP
- 2) Uridine monophosphate UMP
- 3) Thymidine monophosphate TMP.

Unlike the synthesis of purine nucleotide, six membered pyrimidine ring is made first and then attached to ribose phosphate, which is donated by PRPP.

PRECURSORS FOR THE DE NOVO SYNTHESIS OF PYRIMIDINE (FIGURE 19.8):

- Glutamine provides N₃
- Aspartic acid furnishes C₄, C₅, C₆ and N₁
- Carbon dioxide provides C₂.

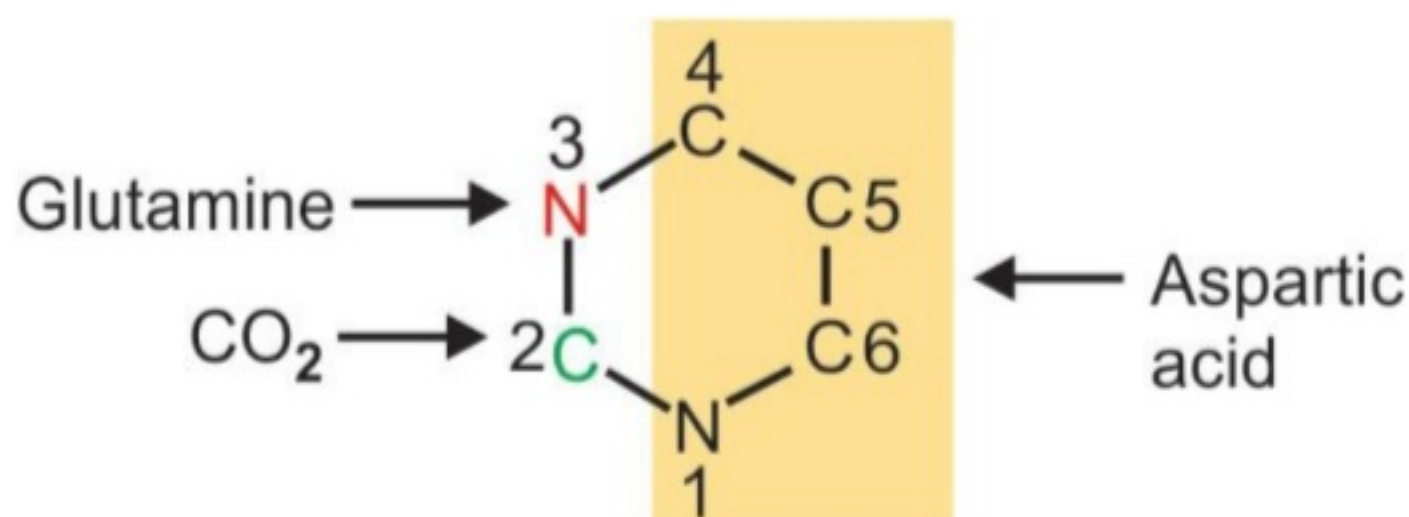


Figure 19.8: Sources of the carbon and nitrogen atoms in the pyrimidine ring

MAJOR STEPS FOR DE NOVO SYNTHESIS OF PYRIMIDINE NUCLEOTIDE:

- 1) Pyrimidine biosynthesis starts with the formation of carbamoyl phosphate from glutamine, ATP and CO₂. This reaction is catalyzed by **cytosolic carbamoyl phosphate synthase-I (CPS-II)** an enzyme different from mitochondrial carbamoyl phosphate synthase-I (CPS-I) required in the synthesis of urea.
- 2) Condensation of carbamoyl phosphate with aspartate forms carbamoyl aspartate in a reaction catalyzed by **aspartate transcarbamoylase** and is the **committed step in the biosynthesis of pyrimidine**.

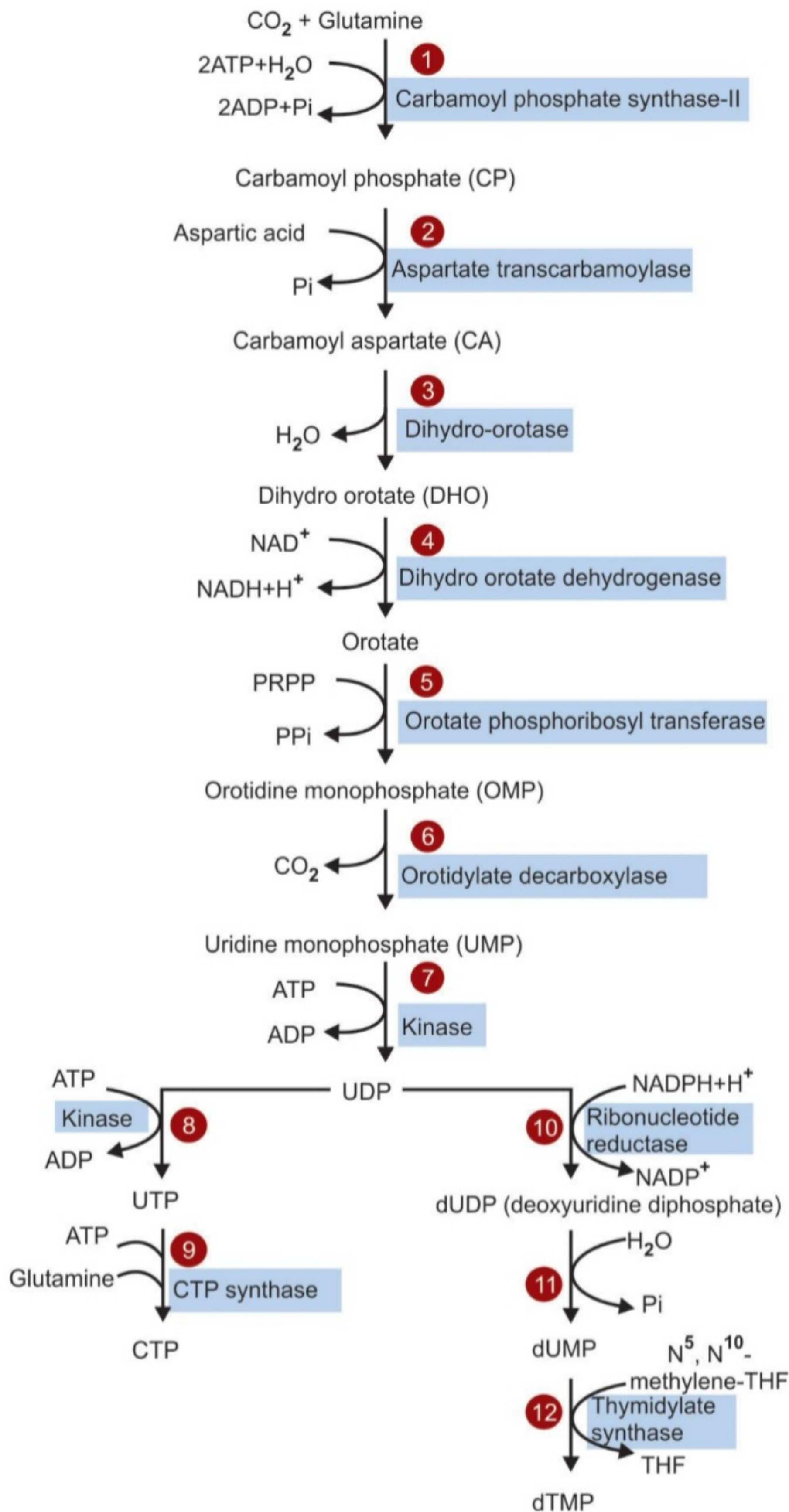


Figure 19.9: De novo pathway for the synthesis of pyrimidine nucleotide

- 3) By removal of water from carbamoyl aspartate, catalyzed by **dihydro-orotase**, the pyrimidine ring is closed with formation of dihydro-orotic acid.
- 4) Dihydro-orotic acid is now oxidized to yield pyrimidine derivative orotic acid, a reaction catalyzed by mitochondrial NAD⁺ dependent **dihydro-orotate dehydrogenase**. All other enzymes of pyrimidine biosynthesis are cytosolic.
- 5) The next step is the acquisition of a ribose phosphate group. Transfer of ribose phosphate moiety from PRPP to orotate, forming orotidine monophosphate, OMP (orotidylate). This reaction is catalyzed by **orotate phosphoribosyl transferase**.
- 6) Orotidylate is then decarboxylated to yield uridylate (UMP).
- 7) UMP is phosphorylated to UDP by kinase using ATP as a phosphate donor. UDP serves as a precursor for the synthesis of UTP, CTP, dUMP and dTMP.
- 8) UDP undergoes an ATP dependent kinase reaction to produce UTP.
- 9) UTP is aminated by accepting amino group from glutamine to form cytidine triphosphate (CTP).
- 10) Ribonucleotide reductase converts UDP to dUDP.
- 11) Deoxyuridine diphosphate (dUDP) is dephosphorylated to dUMP which acts as a substrate for thymidine monophosphate (TMP).
- 12) Methylation of dUMP by N⁵N¹⁰-methylene tetrahydrofolate, catalyzed by **thymidylate synthase**, forms deoxythymidine monophosphate (dTMP).

REGULATION OF DE NOVO SYNTHESIS OF PYRIMIDINE NUCLEOTIDES:

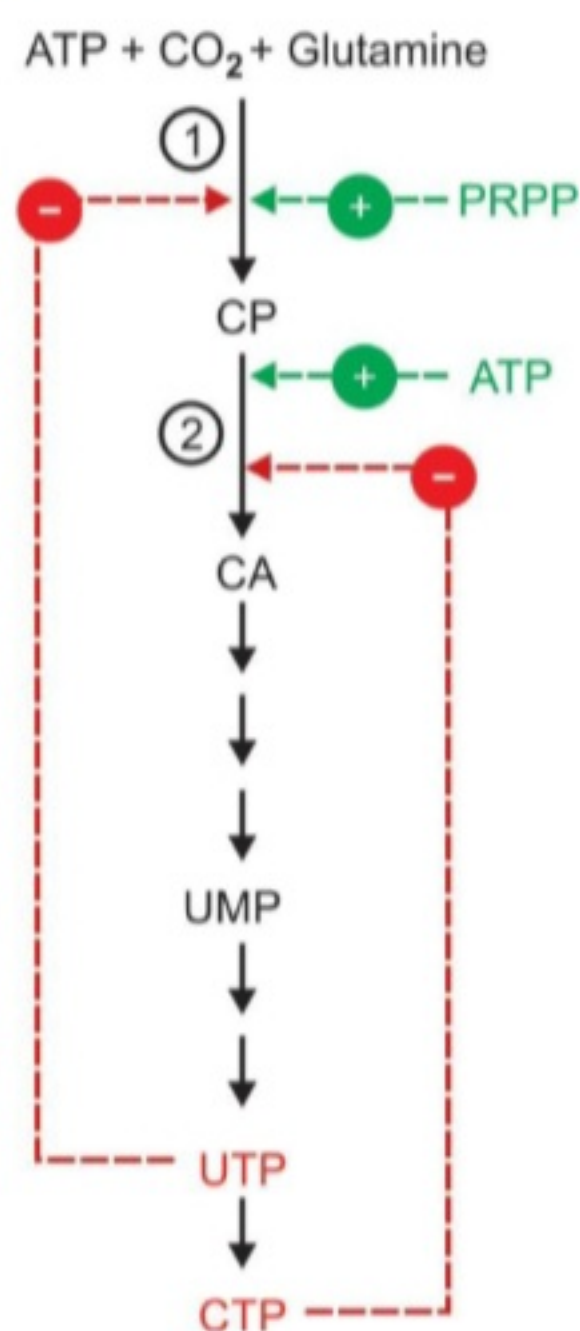


Figure 19.10: Regulation of de novo pyrimidine nucleotide synthesis CP: Carbamoyl phosphate; CA: Carbamoyl aspartate

The first two enzymes **carbamoyl phosphate synthase-II** and **aspartate transcarbamoylase** are allosteric enzymes and are regulated allosterically (Figure 19.10).

- Carbamoyl phosphate synthase-II, reaction 1 is feed- back inhibited by UTP and activated by PRPP.
- Aspartate transcarbamoylase, reaction 2, is feedback inhibited by CTP and activated by ATP.

CATABOLISM OF PYRIMIDINE NUCLEOTIDES (FIGURE 19.11):

Unlike the purines which degraded to sparingly soluble product, uric acid, the end products of pyrimidine catabolism are highly water soluble:

- CO_2
- NH_3
- β -alanine
- β -aminoisobutyrate

Humans probably transaminate β -aminoisobutyrate to methylmalonate semialdehyde which is then converted to succinyl-CoA via methylmalonyl-CoA. β -alanine can serve as precursor of acetyl-CoA.

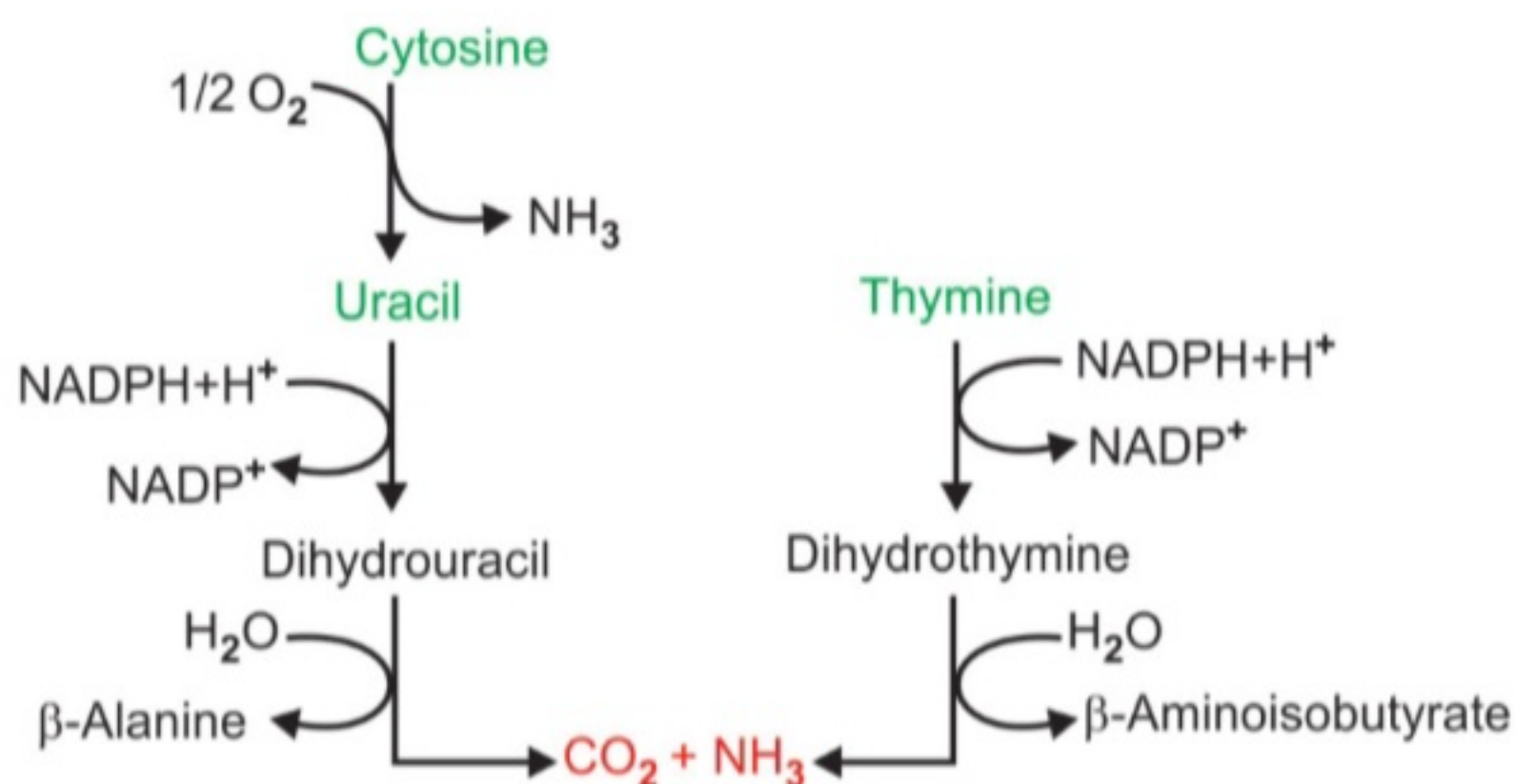


Figure 19.11: Catabolism of pyrimidines